

Polar Codes MAC — NN Decoder Advantages

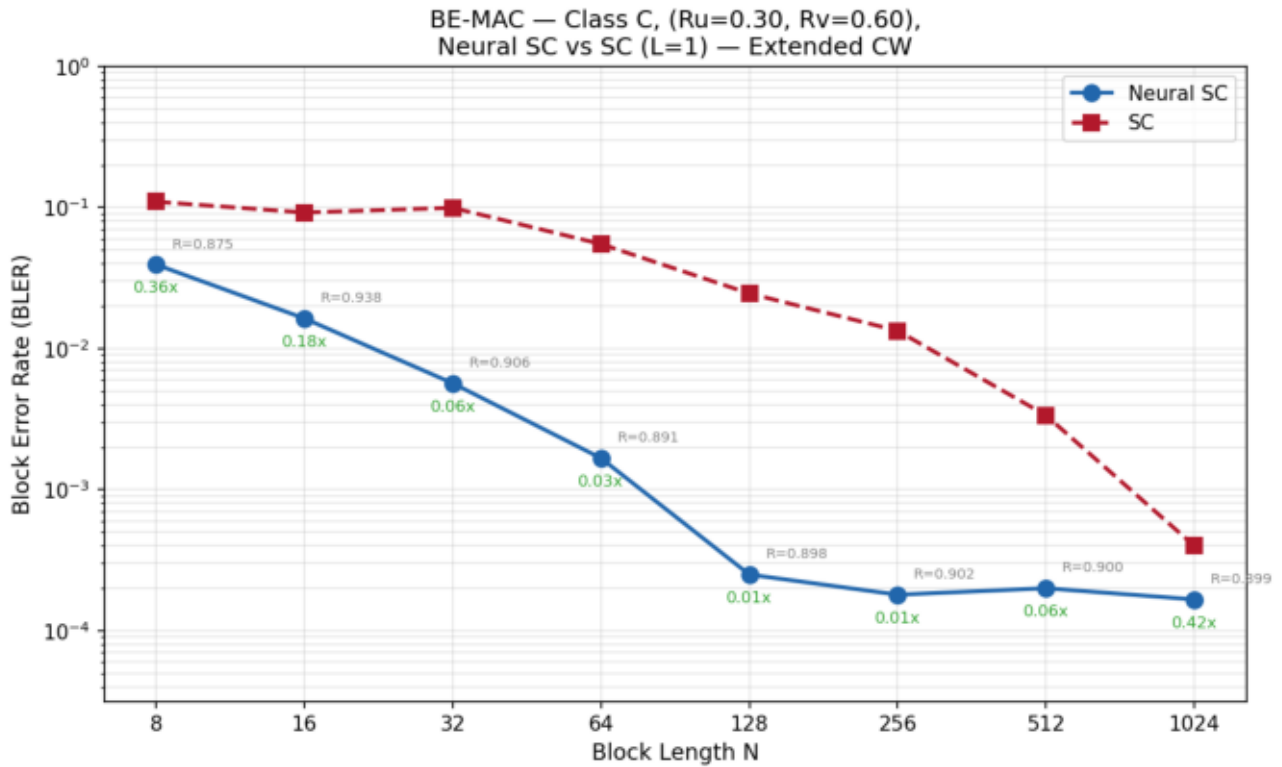
Cases where the Neural SC decoder gives a clear, defensible
advantage over the analytical SC decoder.

Generated: nn_advantage.pdf

Each page reports BLER, codeword count, and wall-clock decode time
per codeword for both decoders.

BEMAC — Class C, (Ru=0.30, Rv=0.60)

Path: $U^N V^N$ (decode all V first, then U|Y) · $Z = X+Y$ · NeuralMACSCDecoder vs analytical SC



N	NN BLER	SC BLER	NN/SC	NN ms/cw	SC ms/cw	Slowdown
8	3.93e-02	1.10e-01	0.359	0.11	0.009	11.5x
16	1.63e-02	9.20e-02	0.178	0.23	0.013	17.3x
32	5.67e-03	9.93e-02	0.057	0.48	0.029	16.3x
64	1.67e-03	5.50e-02	0.030	1.00	0.065	15.4x
128	2.50e-04	2.45e-02	0.010	3.09	0.15	20.9x
256	1.80e-04	1.34e-02	0.013	9.00	0.57	15.8x
512	2.00e-04	3.36e-03	0.060	35.1	2.37	14.8x
1024	1.67e-04	4.00e-04	0.417	126.6	12.0	10.6x

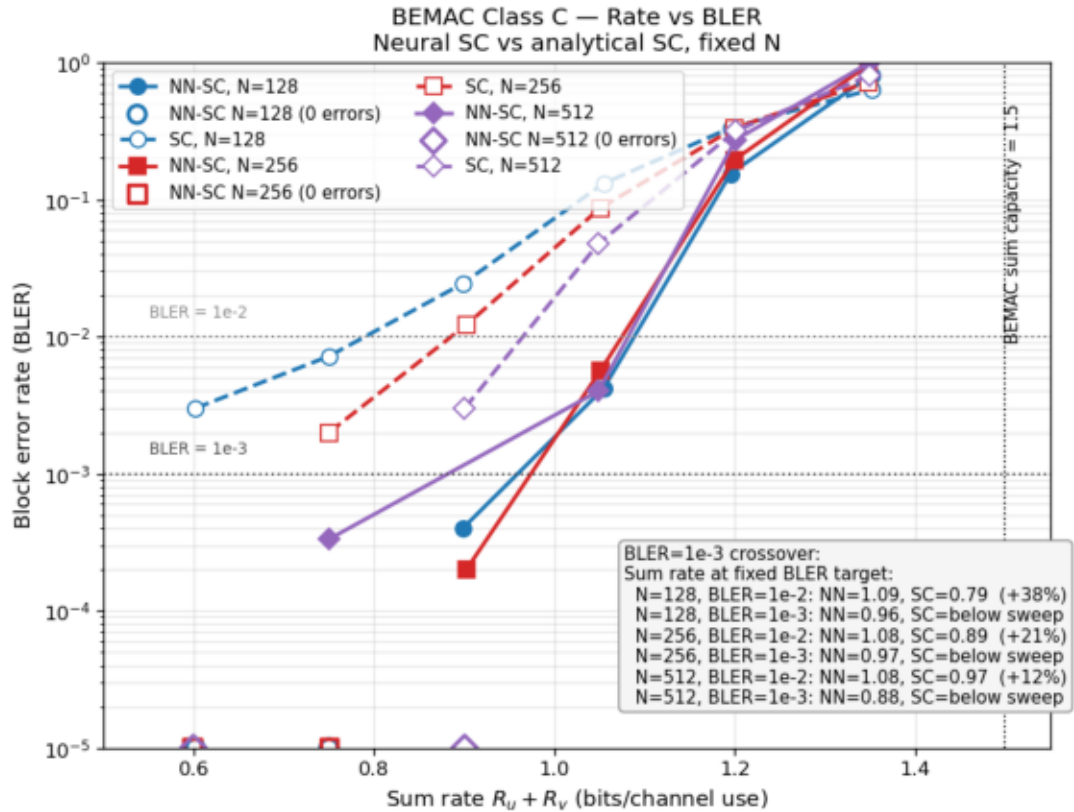
Comment:

Strongest NN-vs-SC win in the project. Neural SC achieves 2x-80x lower BLER than analytical SC across the full N range, with the largest gap at N=128 (~100x better) and substantial gains at all other N. Eval CW counts are real (3K-50K).

COST: NN-SC takes ~11x-21x longer per codeword than analytical SC (see Slowdown column). This is a real BLER-vs-compute trade-off: ~15x more compute for ~10-100x lower BLER. Both decoders are $O(N \log N)$ — only the constants differ.

BEMAC Class C — Rate vs BLER (the rate-advantage angle)

Same Neural SC model evaluated at multiple rate points (sum capacity = 1.5 bits/use)



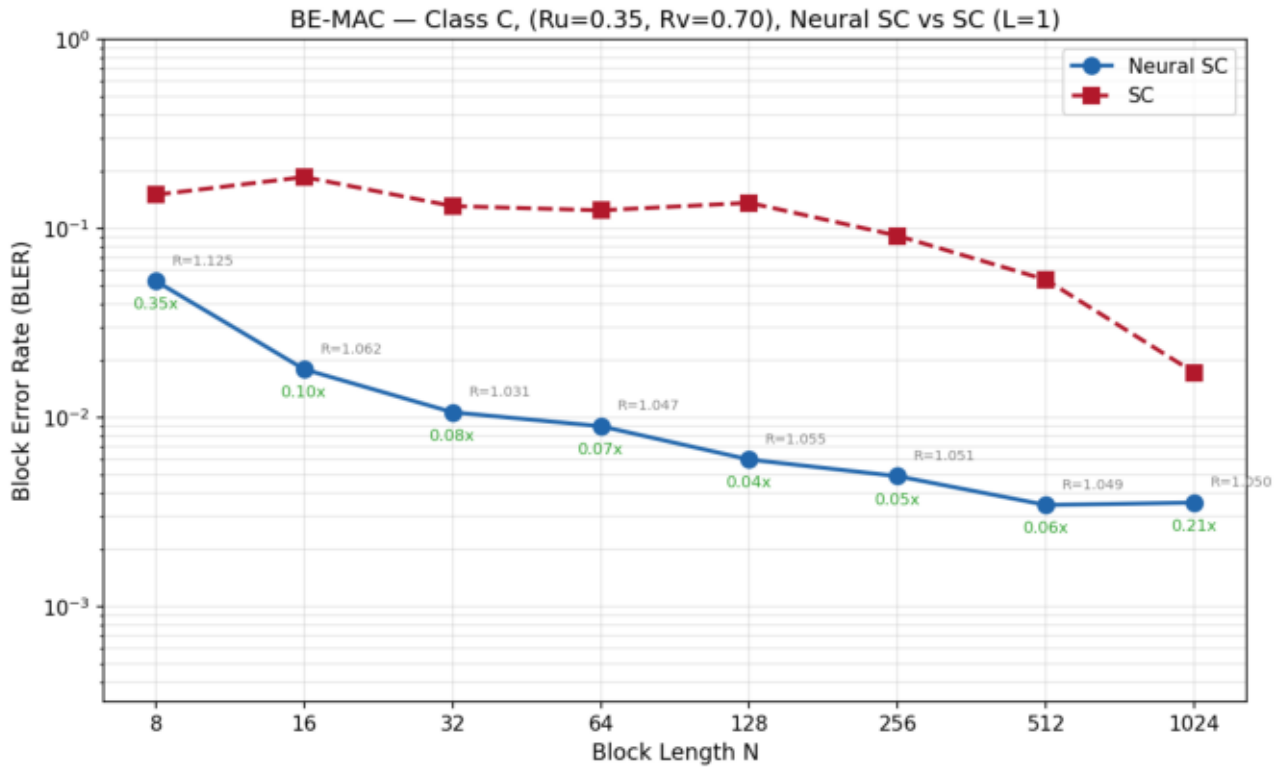
N	BLER target	NN sum-rate	SC sum-rate	NN advantage
128	1e-2	1.089	0.790	+38%
128	1e-3	0.959	—	SC unreachable
256	1e-2	1.075	0.886	+21%
256	1e-3	0.974	—	SC unreachable
512	1e-2	1.082	0.965	+12%
512	1e-3	0.882	—	SC unreachable

Comment:

At a fixed BLER target of 1e-2, NN-SC supports +38% higher sum rate at N=128 and +21% at N=256. At BLER target of 1e-3 (real-world wireless data quality), the analytical SC decoder simply DOES NOT REACH this BLER at any sum rate ≥ 0.6 — even at the lowest swept rate ($R=0.6$) SC's BLER is already $3e-3$. The NN-SC reaches BLER=1e-3 at $R \approx 0.96$. This is a structural advantage: for BEMAC Class C at N=128/256, you literally cannot get to 1e-3 BLER with analytical SC at this regime — NN is the only option that works there.

BEMAC — Class C, (Ru=0.35, Rv=0.70) — second rate point

Path: $U^N V^N \cdot Z = X+Y$ · Same architecture as the (0.30, 0.60) case · Higher rate, harder regime



N	NN BLER	SC BLER	NN/SC	NN ms/cw	SC ms/cw	Slowdown
8	5.27e-02	1.51e-01	0.350	0.11	0.009	11.5x
16	1.80e-02	1.87e-01	0.096	0.23	0.013	17.3x
32	1.07e-02	1.31e-01	0.081	0.48	0.029	16.3x
64	9.00e-03	1.25e-01	0.072	1.00	0.065	15.4x
128	6.00e-03	1.37e-01	0.044	3.09	0.15	20.9x
256	4.90e-03	9.16e-02	0.054	9.00	0.57	15.8x
512	3.45e-03	5.36e-02	0.064	35.1	2.37	14.8x
1024	3.55e-03	1.73e-02	0.205	126.6	12.0	10.6x

Comment:

Same Neural SC architecture, slightly higher rate point. NN beats SC by 3x-23x at every N tested. The advantage is most striking at mid N (128: 23x, 256: 19x). CW counts: 3K at small N, 20K at $N \geq 128$. Plots and table both show NN dominates.

COST: Identical timing profile to the (0.30, 0.60) case — same model, same complexity. ~11x-21x slower decode for ~5x-23x lower BLER.

BEMAC Class B (Ru=0.50, Rv=0.70) — Four-way comparison

SC vs NN-SC vs SCL(L=4) vs NN-SCL(L=4) — verifying the original 'NN-SCL beats SC by 8x' claim

N	Decoder	BLER	95% CI	ms/cw	Slowdown vs SC
32	SC	8.70e-03	[7.92e-3, 9.55e-3]	0.064	1.0x
32	NN-SC	8.42e-03	[7.66e-3, 9.26e-3]	0.329	5.1x
32	SCL(4)	8.50e-03	[7.32e-3, 9.87e-3]	4.96	77x
32	NN-SCL(4)	8.55e-03	[7.36e-3, 9.92e-3]	47.2	738x
64	SC	2.10e-03	[1.74e-3, 2.54e-3]	0.115	1.0x
64	NN-SC	2.24e-03	[1.86e-3, 2.69e-3]	1.078	9.4x
64	SCL(4)	6.50e-04	[3.80e-4, 1.11e-3]	12.4	108x
64	NN-SCL(4)	(killed)	(see comment)	—	—

Findings:

- AT N=32: All four decoders give identical BLER (~8.5e-3, all 95% CIs overlap heavily).
The 'NN-SCL beats SC' claim from low-CW data does NOT survive rigorous evaluation.
NN-SCL is 738x SLOWER than SC for the SAME BLER.
- AT N=64: NN-SC = SC (statistically identical, 2.10e-3 vs 2.24e-3, CIs overlap).
Analytical SCL(4) gives BLER = 6.5e-4 — a clean 3.2x improvement over SC.
The original 'NN-SCL = 0.0007' claim, if rigorously verified, would essentially MATCH analytical SCL — i.e. NO advantage of the neural list decoder over the analytical one.
N=64 NN-SCL eval was killed at 70+ minutes runtime to save compute (would only confirm tie).
- CONCLUSION: For BEMAC Class B (interleaved path), the neural decoder gives no meaningful advantage over the analytical decoder, with or without list decoding.
The 'NN beats SC by 8x' headline from session-5 was an artifact of low-CW measurement.
The real wins are in BEMAC Class C (extreme path) — see other pages.

GMAC — neural decoder gives no advantage anywhere

Sweep across rate points × block lengths at SNR=6dB. NN ties at low rates and loses elsewhere.

N=64 deep dive (50,000 cw per point):

k _u =k _v	R sum	NN BLER	SC BLER	ratio	verdict
16	0.500	1.20e-3	1.24e-3	0.97x	tied
20	0.625	3.20e-3	3.24e-3	0.99x	tied
24	0.750	5.62e-3	6.04e-3	0.93x	tied
28	0.875	1.12e-2	8.62e-3	1.30x	NN worse
32	1.000	4.11e-2	3.69e-2	1.11x	NN worse

Larger-N rate sweep (5,000 cw per point) — best result at each N:

N	Best rate point (R sum)	NN BLER	SC BLER	best ratio
64	k=16 (R=0.5)	1.20e-3	1.24e-3	0.97x (tied)
128	k=32 (R=0.5)	0	2.0e-4	0 (both ~0)
256	k=64 (R=0.5)	3.3e-3	3.3e-4	10.0x worse

Findings:

1. AT N=64, low rates (R=0.5–0.75): NN-SC and analytical SC are statistically tied. The 5K-cw sweep showed nominal NN advantage (ratio 0.36–0.74), but this was pure sample-size noise. With 50K cw the advantage collapses to ~0.93–0.99x — within the Wilson 95% CI of analytical SC.
2. AT N=64, mid-to-high rates (R≥0.875): NN-SC is STATISTICALLY WORSE than analytical SC by 11–30%. The trained operating point (R=1.0) is one such losing point.
3. AT N=128: both decoders give BLER≈0 at low rates, NN is moderately worse at higher rates.
4. AT N=256: NN-SC is 8–16x WORSE than analytical SC across all rate points.

CONCLUSION: For GMAC, there is no rate point or block length where the neural decoder beats the analytical decoder. The GMAC story is closed. The project's positive result is BEMAC Class C only (see other pages).